

Prudential Cleansing Report

Duplicates, dates/currency/percentages, categories and orphan keys — run against the 49,101-row carryover from Stage 3.

Prepared by	Neelam Lahre		
Input	data_clean/_stage3_missing_handled.parquet — 49,101 rows		
Script run	python/02_clean.py (Stages 3-6 appended, single pass)		
Guide reference	Solutioning Guide v1.0, Stages 4, 5 and 6		
Status	Stages 1-6 complete. Stage 7 (SQL star schema) not yet started.		

49,101	2,500	427	46,601
rows entering Stage 4	duplicates removed	date-logic breaches flagged	rows carried to Stage 7

Stage 4 — Exact and Fuzzy Duplicates

Goal: *one policy must appear once.*

Exact duplicates were caught with pandas' row-level **duplicated()**. Fuzzy duplicates — the same customer submitted twice with different formatting — were caught with a fingerprint match key (policy_id + customer_name, stripped to A-Z0-9, joined to a canonical YYYYMMDD date of birth), keeping the most recently updated row per match key as the survivor.

Check	Rows affected	Read
Exact duplicates removed	1,500	Matches the guide's own expected volume exactly
Fuzzy duplicates removed	1,000	Formatting-different resubmissions of the same policy
Total removed	2,500	5.1% of the 49,101 rows entering this stage

Proved: `df.duplicated().sum() == 0` and `df.policy_id.duplicated().sum() == 0` both passed · 46,601 rows remaining · `q_exact_dupes.csv` and `q_fuzzy_dupes.csv` both written to quarantine.

Stage 5 — Dates, Currency and Percentages

Goal: *text becomes typed dates, numbers and one consistent 0-1 scale.*

Currency text ("S\$3,200.00") and mixed-scale percentages (0.284 vs 28.40%) converted cleanly on the first pass — confirmed below. Date parsing did not: the first run produced unparseable rates above 50% on four of five date columns.

Debugging note — a real gap found and fixed

The guide's own **DATE_FORMATS** list includes `%m-%d-%Y` (US month-first) but omits `%d-%m-%Y` (day-first) — the convention this Singapore extract actually uses. Values with a day above 12 (unambiguous, e.g. 25-06-2024) failed every listed format outright, and the unparseable counts below in the first column show the damage before the fix. Adding `%d-%m-%Y` ahead of `%m-%d-%Y` in the format list — so Singapore's convention wins on any ambiguous value — resolved it completely.

Column	Unparseable — before fix	After fix
date_of_birth	26,175	0
quote_date	27,923	0
policy_start_date	27,943	0

lapse_date	2,643	0
claim_date	16,836	0

Business-rule check, post-fix: 427 rows flagged **DATE_LOGIC_BREACH** — a policy starting in the future or before 1990, a quote after its own policy start, or a claim before cover began. This number is only trustworthy now that dates parse correctly; before the fix it would have been a severe undercount, since a null date can never fail a logic check.

Currency & percentage conversion, confirmed: annual_premium_sgd and nbp_margin_pct both hold as float64, policy_start_date as datetime64[ns]. nbp_margin_pct ranges 0.0254 – 0.340 across 46,450 non-null rows — entirely inside the 0–1 scale, confirming the percent-vs-fraction fix held.

Proved: all five date columns show 0 unparseable · dtypes confirmed correct · nbp_margin_pct min/max both within 0-1.

Stage 6 — Categories, Invalid Values, Orphan Keys

Goal: every category means exactly one thing; out-of-range values are flagged, not silently kept.

Category standardisation (typo/casing matched via difflib)

Field	Unmapped after matching	Read
distribution_channel	0	All 34 raw spellings resolved to the 5 real channels
policy_status	0	Fully resolved
claim_status	0	Fully resolved
product_category	173	Expected — seeded junk values (LEGACY-XX, MISC) with no real match

Range validation (RULES dict, Stage 6.3)

Field	Out of range
age (18-100)	367
annual_premium_sgd (1-250,000)	367
sum_assured_sgd (1,000-10,000,000)	0
claim_amount_sgd (1-1,000,000)	236
claim_tat_days (0-365)	220
copay_pct (0-100)	334
nbp_margin_pct (0-1)	0
fraud_risk_score (0-1)	0

Two things worth a second look before Stage 7: age and annual_premium_sgd flag the exact same count (367) — plausibly the same seeded rows carrying two defects at once rather than a coincidence, worth a quick cross-check. And orphan_agent_ids (169) used an adapted rule — no dim_agent_master.csv shipped with this dataset, so any agent_id present without a matching agent_name was treated as unverifiable and redirected to AGT-UNKNOWN, same as the guide's own fallback for a failed reference join. Both are logged, not hidden.

Final row status

dq_row_status	Rows	Share
OK	45,715	98.1%
REVIEW	886	1.9%

Proved: data_clean/policy_clean.parquet and .csv both written (46,601 rows) · logs/dq_report.csv holds one row per check across Stages 3-6 · no rows silently dropped — REVIEW rows stay in the table, excluded only at the KPI-view layer in Stage 9.

Next: Stage 7 — build the SQL star schema (dimensions + fact tables) that data_clean/policy_clean.parquet will load into at Stage 8.